

Smart Fertilizer Spraying System Using Plant Disease Severity Analysis

by

Y Trinesh Reddy

Roll No. 234G5A3311

G Sreeja

Roll No. 224g1A33A9

B Revathi

Roll No. 224G1A3380

Y Sravani

Roll No. 234G5A33A4

Under the guidance of

Dr .C .Sasikala M.Tech ,Ph.D.

Associate Professor



Department of Computer Science and Engineering (AI & ML)

Srinivasa Ramanujan Institute of Technology

(Affiliated to JNTUA & Approved by AICTE) (Accredited by NAAC with 'A' Grade & Accredited by NBA (EEE, ECE & CSE))

Rotarypuram Village, B K Samudram Mandal, Ananthapuramu – 515701.

2025 - 2026

Contents

- Abstract
- Introduction
- Literature survey
- Proposed Work
- References
- GitHub Link
- Queries



Abstract

The Smart Fertilizer Spraying System Using Plant Disease Severity Analysis integrates Artificial Intelligence (AI) and Internet of Things (IoT) technologies to optimize crop management and improve agricultural productivity. High-resolution images of crop leaves are captured through IoT-enabled cameras and analyzed using deep learning models to identify disease symptoms and determine severity levels. Based on the detected severity, a microcontroller-controlled spraying unit automatically adjusts the type and quantity of fertilizer or pesticide, ensuring precise and need-based application. This approach minimizes chemical wastage, reduces costs, and lowers environmental impact while enhancing plant health and yield. Real-time data monitoring through a cloud dashboard allows farmers to track crop conditions, receive alerts, and make timely decisions. The system promotes sustainable agriculture by combining automated disease detection with intelligent resource management.



Introduction

- Agriculture is vital for food security and economic stability but faces challenges like crop diseases, chemical overuse, and inefficient resource management.
- Manual disease detection is slow and often inaccurate, while excessive fertilizer and pesticide use raises costs and harms soil and the environment.
- The Smart Fertilizer Spraying System Using Plant Disease Severity Analysis uses AI and IoT to automate disease detection and fertilizer application.
- IoT cameras capture leaf images, machine-learning models identify disease type and severity, and a microcontroller controls a sprayer to apply the exact amount of fertilizer or pesticide.
- This data-driven approach cuts chemical waste, lowers labor, improves yield, and promotes sustainable farming by enabling early detection and precise treatment.



Literature Survey

No	Title	Author	Journal Name & Year	Methodology Adapted	Key Findings	Gaps
1.	Using Deep Learning for Image-Based Plant Disease Detection.	S. Mohanty, D. Hughes, MSalathé	Frontiers in Plant Science, 2016	CNN-based image classification for disease detection	CNN models gave >90% accuracy for plant disease detection, proving automated image-based diagnosis is feasible.	Limited to publicly available datasets, may not generalize to all crop types
2	Deep Learning in Agriculture: A Survey	A. Kamilaris, F. X. Prenafeta-Boldú	Computers and Electronics in Agriculture, 2018	Surveyed CNN, RNN, and IoT methods for smart farming	Reviewed AI methods (CNN, RNN, IoT) for smart farming; stressed real-time and scalable solutions.	Mostly theoretical, lacks practical IoT implementation examples
3	IoT-Enabled Smart Farming System for Disease Monitoring and Fertilizer Control	H. Zhang, W. Li, S. Yang	IEEE Access, 2020	IoT sensors and microcontrollers used for automated spraying based on sensor input	IoT-based system reduced fertilizer waste and improved disease response time in small-scale trials.	Limited to small-scale fields, cloud integration not fully explored

Proposed System

- Our proposed work presents an AI–IoT based smart system for detecting plant diseases and automatically controlling fertilizer or pesticide spraying.
- By capturing real-time leaf images through IoT cameras and analyzing them with a CNN-based model, the system identifies disease type and calculates severity.
- A microcontroller processes the severity data and regulates a motorized sprayer to apply the exact quantity of fertilizer or pesticide required.
- A cloud dashboard with mobile alerts allows farmers to monitor field conditions and receive timely notifications, ensuring precise treatment, reduced chemical usage, and improved crop yield

Reference

- [1]. S. Mohanty, D. Hughes, and M. Salathé, "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science*, vol. 7, no. 1419, pp. 1-10, Oct. 2016.
- [2]. A. Kamilaris and F. X. Prenafeta-Boldú, "Deep Learning in Agriculture: A Survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70-90, May 2018.
- [3]. L. Li, J. Zhang, and Z. Huang, "A Review of IoT-Based Smart Agriculture: Architectures, Applications, and Challenges," *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 8171-8184, Oct. 2019.
- [4]. M. Ferentinos, "Deep Learning Models for Plant Disease Detection and Diagnosis," *Computers and Electronics in Agriculture*, vol. 145, pp. 311-318, Jan. 2018.

Git Hub Dashboards of each student

The screenshot shows the GitHub interface for a repository named 'CSM-2025-26-Batch-B-10' by user 'SreejaGanne'. The repository is public and has 0 stars, 0 forks, and 0 watches. The main branch is 'main'. The repository contains two files: 'README.md' (initial commit, 46 minutes ago) and 'Research Paper 1.pdf' (added via upload, 14 minutes ago). The README file is selected, showing the title 'CSM-2025-26-Batch-B-10'. The right sidebar shows the 'About' section with no description, website, or topics provided. Below 'About' are links for 'Readme', 'Activity', '0 stars', '0 watching', and '0 forks'. The 'Releases' section shows 'No releases published' with a link to 'Create a new release'. The 'Packages' section shows 'No packages published' with a link to 'Publish your first package'.

SreejaGanne / CSM-2025-26-Batch-B-10

Type / to search

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

CSM-2025-26-Batch-B-10 Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file <> Code

SreejaGanne Add files via upload c99f83e · 14 minutes ago 2 Commits

README.md	Initial commit	46 minutes ago
Research Paper 1.pdf	Add files via upload	14 minutes ago

README

CSM-2025-26-Batch-B-10

About

No description, website, or topics provided.

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Any Queries?

*Thank
You!!!*